

# Harith Al-Safi

Machine Learning Engineer

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## 📄 About me

An experienced Full Stack Engineer with **3+ years** of specializing in Machine Learning, React TypeScript, Java Spring Boot, and Cloud Technologies. Proven record in leading technical teams, and driving cost optimizations across **startup** to **enterprise**.

## 📁 Experience

### Full Stack Engineer

📅 Sep 2024 - Ongoing (12+ Months)

BT Group

📍 Manchester, UK

- Delivered 5 **Kafka-based Spring Boot microservices** with MongoDB/PostgreSQL, Netflix Conductor, and Pulumi
- Mentored and examined 5+ apprentices on agile development with Java Spring Boot and **AWS Lambda**
- Achieved **30% AWS cost reduction** by automating infrastructure with TypeScript and optimizing ECS cluster
- Re-architecture Spring Boot APIs with JWT, Azure AD, and **role-based access** achieving **enterprise grade security**
- Led **React.js to Next.js** migration of 5K+ lines, adding Cypress testing and OpenAPI to TypeScript **code gen**
- Managed regional Graduate Network organizing and coordinating monthly tech talks for **30+ attendees**

### Robot Engineer

📅 Dec 2023 - Mar 2024 (4 Months)

Robot Fight League

📍 Leeds, UK

- Designed a 3D CAD for robot fight league using Solidworks, Blender and Jira for project management
- Implemented a control system using Arduino and C++ for the robot's movement, sensors and pneumatic system
- Tested object detection and tracking using OpenCV and Python for the robot's vision system
- Managed a team of 5 engineers to develop the robot and compete in the league ending up 1<sup>st</sup> place

### Software Engineer (Placement Year)

📅 July 2022 - July 2023 (12 Months)

Johnson Controls

📍 London UK

- Invented a **custom scripting language with a compiler** for C# app increasing development productivity by 40%
- Managed a team of 10 interns to create a **business plan** and ended up placing 2<sup>nd</sup> amongst the whole company
- Administered C/C++ projects while fixing bugs, adding features, refactoring and **migrating 90% of codebase to C#**
- Developed C# .NET libraries for fire detection simulations and data analysis improving operations by 32%
- Collaborated on an **analytics web dashboard with React.js** for predictive maintenance with dev ops and CI/CD
- Maintained Python automation scripts for automated builds of .NET applications and processing data with Azure

### Machine Learning Engineer Freelance

📅 Dec 2021 - Mar 2022 (4 Months)

Paperound

📍 Leeds UK

- Applied data visualization using Python Matplotlib to present analytical insights to non-technical stakeholders
- Created natural language processing applications using LLM's and Python for sentiment analysis and chatbots
- Utilized Python, TensorFlow, and Scikit-learn to build, train, and evaluate neural networks
- Built and trained a reinforcement neural network using TensorFlow, KerasRL for predicting stock prices
- Developed distributed system using C++ OpenMP, OpenCL and CUDA for parallel statistical computing
- Engineered a high performance C++ linear algebra library for data science through Jenkins, Docker and Cmake

### IT administrator

📅 Jun 2020 - Mar 2021 (10 Months)

Mesopotamia Group

📍 Amman Jordan

- Architected **scalable file server infrastructure** using Apache, Nextcloud, and Ubuntu in production environment
- Automated system administration workflows using **Python, Bash scripting**, and network security protocols
- Developed **high-availability load balancer with C/MPI** achieving 34% improved system uptime
- Built **predictive analytics pipeline using Python** for proactive cybersecurity threat detection
- Designed cloud networking solutions implementing TCP/IP, routing protocols, and infrastructure optimization

## 📖 Education

BEng (Honours) in Electronics and Computer

📅 Sept 2019 - July 2024

- Cloud and Parallel computing, Secure and Distributed systems, Cybersecurity, Software Engineering, Data Science
- Embedded systems, Microprocessors, FPGA, Circuit Analysis, Signal processing, Networking, Compiler design

Skills and Languages

Technical:

C \ C++PythonJavaC#JavaScriptLinuxDockerAzureAWSJiraGitJenkinsTerraformsKubernetesSQLiteMySQLMongoDBPostgreSQL.NET WPFJavaFxReact JsQT FrameworkFlutterFlaskSpringbootASP.NETNumPyKerasRLMatplotlibPytorchSciPyTensorFlowPandasOpenCvCMakeNode.jsOpenCLOpenGLOpenMPCUDAArm Mbed OsArm KeilFPGAVerilog HDLArduinoBlenderPhotoshopAutoCAD

Soft:

ManagementLeadershipMotivatedDeterministicCommunicationProblem solver

Languages: Arabic (Full Professional)

English (Full Professional)

Spanish (Elementary)

Projects

MCPP: High Performance C++ Library for Numerical Computation

Apr 2021 - Oct 2022

Personal

- Created a cross-platform math library for C++ used for matrix algebra, statistics and machine learning.
- Engineered with CI/CD using Jenkins, unit tests, Docker, Cmake, Doxygen and Git for version control
- Improved performance by 40% using parallel computing with OpenMP, OpenCL and CUDA

TBot: Machine Learning Stocks Trading Bot

Apr 2021 - Oct 2021

Personal

- Developed gradient boosting machine learning model using Python and Scikit-learn for stock prediction
- Anlayzed and processed data using Pandas, Numpy and Scipy for training and testing the model
- Backtested the model using historical data and simulated trading using Python and Matplotlib
- Developed a REST API and a user interface to interact with the bot using AWS, Flask and PyQT

Vega: LLM Chat Web App for IoT Circuits

Oct 2023 - Jun 2024

UoL

- Developed a full stack chat bot web application with Typescript, React.js, Next.js, TailwindCSS and NoSQL
- Programmed natural language processing algorithms using LLMs to interpet and execute user messages
- Developed a Python Flask REST Server with SQLite on the Raspberry Pi to communicate with the chat bot
- Engineered a circuit with sensors, fans, LCD and buttons to execute the user's commands from the chat bot

Professional Courses

Data Analysis with Python

Oct 2021

- Develop Python code for cleaning and preparing data for analysis
- Perform exploratory data analysis and analytical techniques to datasets using Pandas, Numpy and Scipy
- Manipulate data using dataframes, understand data distribution, perform correlation and create data pipelines
- Build and evaluate regression models using machine learning scikit-learn for prediction and decision making

Databases and SQL for Data Science with Python

Sept 2021

- Analyze data within a relational database on the cloud using IBM DB2, SQL, Python
- Construct SQL CRUD operations and compose views, transactions, stored procedures and joins.

Python for Data Science, AI & Development

Aug 2021

- Demonstrate proficiency in using Python libraries such as Pandas, Numpy, and Beautiful Soup
- Access web data using APIs and web scraping from Python in Jupyter Notebooks.

Extracurricular

AI Agent hackathon

Mar 2023

Lablab

Remote

Web Developer hackathon

Feb 2023

**Event Coordinator Volunteer**  
IEE RAS ICRA

📅 Jan - 2023  
📍 London, UK